

SMART STUDY PLANNER – AN ACADEMIC PRODUCTIVITY SYSTEM

PROJECT DESCRIPTION DOCUMENT

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1. INTRODUCTION

Smart Study Planner is an intelligent and adaptive academic management system designed to help students effectively organize their studies while considering their mental well-being. Traditional study planners primarily focus on task scheduling and deadline tracking, but they fail to incorporate critical real-life factors such as fatigue, stress levels, sleep quality, and motivation. This system addresses that gap by integrating both academic and psychological parameters into a unified planning solution. In modern academic environments, students are required to manage multiple subjects, assignments, and deadlines simultaneously. This often leads to poor planning, burnout, and reduced productivity due to the lack of awareness of their actual working capacity. Smart Study Planner introduces a more realistic approach by combining task management with mental load analysis, allowing students to generate schedules that align with their current physical and mental condition.

Overall, Smart Study Planner functions as both a scheduling tool and a decision-support system, helping users make informed choices about their study routines while maintaining a healthy balance between productivity and well-being.

2. OBJECTIVE

The primary objective of the Smart Study Planner is to develop a system that assists students in managing their academic workload efficiently while taking into account their mental state. The goal is to move beyond traditional scheduling methods and introduce a more intelligent, adaptive, and user-centric approach to study planning.

Another important objective is to design a secure and robust full-stack application that ensures reliability and data protection. The system implements several security measures, including OTP-based user registration, password reset functionality, JWT-based authentication, middleware validation, centralized error handling, and rate limiting.

3. KEY FEATURES

- **User Authentication System**
Secure login and registration using OTP verification, JWT tokens, and encrypted passwords
- **Subject and Task Management**
Full CRUD operations for subjects and tasks with attributes like deadlines, priority, difficulty, and estimated time
- **Mental Load Tracking**
Recording of fatigue, sleep, and stress-related parameters to evaluate the student's current condition
- **Automated Schedule Generation**
Intelligent scheduling based on task priority, deadlines, and mental capacity
- **Analytics and Insights**
Provides data-driven insights such as productivity score, fatigue trends, and burnout probability
- **Partial Frontend Implementation**
User interfaces for login, dashboard, subjects, tasks, mental load, and analytics

These features work together to reduce manual planning effort, improve consistency, and provide a more holistic approach to academic management.

4. TECHNOLOGY STACK

The Smart Study Planner is developed using a modern full-stack technology stack that emphasizes strong backend architecture and overall system reliability. The frontend is built using React.js with Vite, enabling fast performance and a smooth user experience, while React Router is used for efficient navigation between different pages and Axios facilitates seamless communication with backend APIs. On the backend, the system is powered by Node.js and Express.js, which handle server-side logic and API management effectively. For data storage, MongoDB is used along with Mongoose ORM, allowing structured and flexible handling of application data.

In terms of security and supporting utilities, the system incorporates JWT (JSON Web Tokens) for secure authentication and session management, bcrypt for password hashing to ensure data protection, and Nodemailer for implementing email-based OTP verification. Additionally, Swagger is used for API documentation, making the system easier to understand and maintain, while Jest and Supertest are utilized for testing to ensure reliability and performance. Overall, the system follows a backend-first approach, ensuring that all critical functionalities such as authentication, scheduling, and analytics are robust, scalable, and fully implemented.

5. SYSTEM ARCHITECTURE

The Smart Study Planner follows a full-stack layered architecture where the frontend interacts with the backend through RESTful APIs, and the backend manages business logic, data processing, and database operations. Users create subjects and tasks, provide mental load inputs such as fatigue and sleep, and the system processes this data using scheduling algorithms to generate a personalized study plan. It also computes analytics based on past data to provide insights. Secure authentication and protected routes ensure that users can only access their own data, while the modular design enhances scalability, maintainability, and overall performance.

6. SECURITY AND DATA HANDLING

Security is a critical component of the Smart Study Planner, with multiple layers implemented to ensure safe and reliable data handling. The system includes OTP-based authentication for secure registration, JWT-based session management, and password encryption using bcrypt.

Additional safeguards such as middleware validation, centralized error handling, and rate limiting help maintain data integrity and prevent misuse of APIs. All user data is securely stored, and protected routes ensure that users can only access their own information, preserving privacy and security.

7. FUNCTIONAL OVERVIEW

Smart Study Planner provides a unified platform where users can manage subjects, tasks, and mental load through a centralized dashboard. It generates personalized study schedules by analyzing factors like deadlines, task difficulty, fatigue, and available capacity using a heuristic-based approach.

Key concepts such as Task Priority Score and fatigue-based capacity adjustment ensure realistic and efficient planning. The system also offers insights like productivity scores, fatigue trends, and burnout probability, helping students improve their study habits over time.

8. ORIGINALITY STATEMENT

This project is an original contribution to academic planning systems, as it integrates mental health factors such as fatigue, stress, and motivation into the scheduling process. Unlike traditional planners, it combines multiple parameters into a single decision-making engine to generate personalized and realistic schedules. The use of a heuristic-based framework ensures transparency and interpretability, while the implementation of modular architecture, secure authentication, API documentation, and testing reflects strong software engineering practices.